

Space age

A secret alien report that has finally been declassified. And we've got a copy

**3D Guns Illegal**

Guns were always illegal and now 3D printing of weapons or having blueprints is illegal too in Australia <http://dgit.in/3Dgunsban>

PHYSICS (AND A LITTLE TECH) OF STAR WARS

Ronak Gupta
readersletters@digit.in

With the new installment of the Star wars franchise slated to release December 18, most of us would probably want to forward time and watch it as soon as possible. That John Williams score, freaking trailers and the general magnitude of what Force Awakens is likely to unveil gets us geeks jumpy to our core. In this article we aim to see Star Wars holds up when parsed through the filter of current day physics. We dare not be critical of something that is almost sacred to millions. But if not analytical, what else do you expect true geeks to be?

Tatooine : Fact or fiction?

One of the most iconic shots from the entire Star Wars saga is the one where Luke Skywalker stands on his home planet Tatooine and watches twin suns setting. Tatooine unlike Earth orbits around not one but two stars. It's an awesome shot, sure, but does physics allow for a planet to orbit around a binary star system? According to recent studies and evidence it does. A Tatooine sunset is possible after all. For a long time it was believed that the combined gravitational force exerted by two suns would provide too hostile an environment for an orbit of a rocky planet and Tatooine might just be a figment of a creative genius's imagination. In 2011 however,

We analyse what they got right and what they got wrong. But beyond that we also tell you about current-day technologies that could one day make those great leaps of imagination actually come true.

scientists discovered a half gas and half rock-ice planet dubbed Kepler-16(AB)-b that orbits the binary star system Kepler 16-b. But how do the planets form midst all that gravity exerted by the twin suns? It's simple. They don't. According to a recent study a planet actually forms away from the binary star center and then gradually settles into an orbit. Also if the binary stars are extremely close together, a planet such as Tatooine can be in orbit far away from the twins; so far away in fact that the planet practically experiences the gravitational field of the twins as one. So... Tatooine can exist, but can it support life? So far though it looks like our luck has maxed out as most

real planets orbiting binary suns are cold and inhospitable as we know it. Tatooine as depicted in the movies however is largely a desert world which in theory can support life. But, there has never been a hint

of rain or the presence of an oasis on the planet, so how do the natives of Tatooine survive in this arid, dry wasteland? Advancement in technology has now made it possible (if only in the 'foreseeable' future) to harvest water from moisture and indeed moisture farmers make their appearances in the movies. Astrobiology questions aside it is pretty cool that Star Wars foreshadowed the existence of a planet with twin suns – an amazing scientific discovery.

Death Star : Futuristic weapon or something out of a megalomaniac's dream?

Even people not familiar with Star Wars surely would've heard of the Death Star. In an epic scene from the movie franchise the entire planet of Alderaan is blown apart to smithereens by a single ray from the Death Star. Alderaan is not a marble, or a dwarf planet. By all estimates, it is a solid Earth sized planet. How much energy does it supposedly take to destroy an entire planet? Does current technology come close to replicating the Death Star's power.

A few back of the envelope calculations suggest that if the Death Star harnessed our Sun's nuclear fusion produced energy,